Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/01/2022
PI Name:	Tandon, Ritesh Ph.D.		
Project Title:	Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	rtandon@umc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	601-984-1705
Organization Name:	University of Mississippi Medical Center		
PI Address 1:	Department of Microbiology and Immunology		
PI Address 2:	2500 N State St		
PI Web Page:			
City:	Jackson	**State:**	MS
Zip Code:	39216-4500	**Congressional District:**	3
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/01/2020	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:	NOTE: P.I. changed to S. Shay effective 10/05/2023. For subsequent reporting, please see "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication (P.I.-- Shay)". (Ed. 10/15/23)		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Mehta, Satish Ph.D. (NASA Johnson Space Center) Simpson, Richard Ph.D. (University of Arizona)		
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Performance Goal Text:			

Task Description:	Opportunistic pathogens such as latent viruses pose a unique hazard to crew members because of the ineffectiveness of preflight quarantine periods and decreased immune function. Latent herpesviruses reactivate and shed in higher quantities during and after long and short spaceflights. Human cytomegalovirus (CMV) is an important herpesvirus from a public health viewpoint that causes severe systemic infections upon primary infection or on reactivation from latency. The few antiviral drugs available to treat CMV infection suffer from safety, efficacy, and resistance issues, and no vaccine against CMV is currently available. As NASA and collaborating space agencies plan for future exploration class missions (i.e., to Mars or an asteroid), eliminating the risk of an adverse health event due to alterations in virus latency, reactivation, and lytic replication is paramount. Evidence of impaired control of latent viral infections has been documented during short (i.e., 14-day Space Shuttle missions) as well as long (i.e., 6-month International Space Station missions) duration spaceflight missions, with the magnitude of this response being positively associated with time in flight. Shedding of herpesviruses including Epstein–Barr virus (EBV), varicella-zoster virus (VZV), and CMV was detected in >30% of astronauts in a recent study (Mehta et al., 2017) looking at long-duration spaceflight (60–180 days). These viruses reactivated independently of each other and reactivation frequency increased in duration, and amplitude (viral copy numbers) when compared to short duration (10 to 16 days) space shuttle missions. Thus, it is easy to speculate that the substantially longer durations being proposed for exploration class missions (i.e., up to 3 years) will further increase the risk of an adverse health event due to increased virus reactivation and/or replication. Add increased radiation exposure (both duration and magnitude), limited clinical care, and the impossibility of rapidly returning a casualty back to Earth to the list of potential hazards and it is clear to see why obtaining a better understanding of virus replication under space-like conditions is critical to successfully and safely send humans to, and return them from, these remote extraterrestrial destinations. In this proposal, we aim to determine the impact of space radiation on CMV latency, reactivation, and lytic replication. While most studies on this topic in the past have focused on changes in the human immune system that could induce virus reactivation, we will approach this topic from both virus and host standpoints. We will study differences in CMV latency and reactivation before and after radiation exposure and determine if these changes can be correlated with specific changes in virus genome and/or transcriptome. Also, because virus control, dissemination, and resolution are ultimately governed by the competency, or lack thereof, of the host immune system, we will determine the impact of pre-exposing NK-cells and T-cells to space radiation on their ability to control active CMV infections. In this project, we will focus on a single variable: space radiation, rather than looking at multiple variables such as microgravity, physical and psychological stresses of space travel, diet, medications, or other factors associated with spaceflight. The overarching hypothesis is that modeled space radiation will induce specific changes in CMV genome/transcriptome or immune control of infection that will induce CMV reactivation and/or increase in virus replication and pathogenicity. The specific aims to test this hypothesis are: Specific Aim 1: Determine the impact of space radiation on CMV genome and transcriptome. Specific Aim 2: Correlate radiation induced changes in CMV genome and/or transcriptome with CMV latency, reactivation, and lytic replication. Specific Aim 3: Determine the impact of space radiation on immune cell control in a mouse model of CMV infection. Reference: Mehta SK, Laudenslager ML, Stowe RP, Crucian BE, Feiveson AH, Sams CF, Pierson DL. (2017) Latent virus reactivation in astronauts on the international space station, NPJ Microgravity 3, 11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of this project, we will gain a better understanding of how radiation impacts CMV latency, reactivation, and lytic replication. By measuring gene expression on infected cells exposed to radiation, using both traditional and cutting-edge techniques, we will be able to identify CMV genomic and transcriptomic changes that affect virus behavior. We will also determine if space radiation can impair NK-cell and T-cell function, allowing for CMV to replicate and disseminate in the host. This will allow us to determine the risk of CMV infection during spaceflight missions and gain a better understanding of the potential underpinning mechanisms. Findings from this study could lead to the development of appropriate countermeasures to ameliorate potential clinical risks associated with CMV reactivation in flight. Moreover, the methods incorporated here can be adapted to examine the effects of radiation on other relevant viral pathogens known to cause significant disease in immunocompromised people that may also pose a risk to crewmembers during spaceflight missions. Discovering the mechanisms of virus reactivation will help with the development of treatment modalities for patients on Earth.
	OCTOBER 2023 UPDATE NOTE: P.I. changed to S. Shay effective 10/05/2023. For subsequent reporting, please see "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication (P.I.-- Shay)". (Ed. 10/15/23) ### Kasumi cells are hematopoetic stem cell derived monocyte progenitor cells that were infected with a low dose of cytomegalovirus (CMV) and establishment of latency was monitored by analyzing viral gene expression as well as genome copy numbers. Kasumi cells offer an established model of CMV latency and are amenable to latency establishment as well as reactivation in the laboratory. They do not allow lytic viral replication unless differentiated. Once CMV latency was established in these cells, the cells were exposed to gamma, proton, and galactic cosmic radiation (GCR) radiation (0.1, 0.25, 0.5, and 1.0Gy) for different time points (e g., 0 to 30 days) at the Brookvaven National Laboratory (BNL) in New York. The cells were harvested at different times after the irradiation and analyzed for viral load by real-time polymerase chain reaction (PCR). It was interesting to note that there was an increase in the viral load for almost all the radiation concentrations used six days after the exposure of gamma, proton, and GCR. The data is being analyzed for statistical significance and interpretation. To date, we have performed 3 out of our proposed 5 ion experiments (during the Spring and Summer 2021 sessions) to determine the impact of space radiation on CMV genome, transcriptome, and reactivation potential. Data from those experiments have been generated and are being analyzed. The addition of the Spring/Summer 2023 experiments will close this aspect of the study and allow for continuation into the next in vivo experimental phase of the project. The Principal Investigator (PI) and team’s accelerator experience: Dr. Satish Mehta has been to BNL on numerous previous occasions and assisted in the ion radiation exposure of our latently infected Epstein-Barr Virus cell lines. Dr. Douglass Diak has never been onsite at BNL but organized the Spring and Summer 2021 shipping/receiving of the

	latently infected CMV cell lines to/from BNL. In the process, he completed all necessary safety training for onsite BNL work as well.
Task Progress:	We conducted a similar study at NSRL and the current beam times requested match our historical times, but more importantly have been established using the NSRL beam time calculator. The exposure times are in direct relation to the energy dose requirements (10, 50, 100, and 200cGy), which again match our past radiation studies. Our radiation expert here at NASA Johnson Space Center (JSC) has recommended these energy doses as well as these radiation sources: gamma, proton, C-12, and Fe-56-. His suggestions are based on the probability of exposure to these sources during actual spaceflight, making our radiation source selection highly relevant to our space study correlations. Though we understand that cosmic radiation is a combination of these individual sources, we believe our understanding of CMV reactivation in response to each of these sources, independently, is crucial. Again, these sources and doses were recommended to us by our resident radiation expert. Additionally, we conducted a similar study and found that these sources and doses were highly effective in stimulating Epstein-Barr Virus (EBV) to reactivate. See: Reactivation of latent Epstein-Barr Virus: A comparison after exposure to gamma, proton, carbon and iron radiation, doi:10.3990/ijms19102961. Kasumi cells from the above experiments were also subjected to genomics and transcriptomics studies to reveal single nucleotide polymorphisms or other variations in the genome of irradiated cells compared to mock, and reveal any transcripts that are significantly altered in expression. Total nucleic acid (TNA) was extracted from the Kasumi cells from the above experiment and was sent for DNA as well as RNA sequencing at Microbial genome sequence center (MiGS) in Pittsburgh, PA. Selected samples were sequenced from the above experiment. Genomic analysis: Initial set of 22 samples (pre-radiation, D6, D12, GCR, Proton) did not reveal any genomic variants. The analysis is still ongoing. Transcriptomic analysis: Differentially expressed genes comparing the GCR-Day6 cells to pre-exposed cells were analyzed. Interestingly, only UL49 shows a significantly elevated level in GCR-Day6 cells compared to mock. UL49 mutants are currently being made in the human cytomegalovirus (HCMV) genome in order to study its role in latency reactivation.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Dwivedi R, Sharma P, Farrag M, Kim SB, Fassero LA, Tandon R, Pomin VH. "Inhibition of SARS-CoV-2 wild-type (Wuhan-Hu-1) and Delta (B.1.617.2) strains by marine sulfated glycans." Glycobiology. 2022 Sep 19;32(10):849-54. https://doi.org/10.1093/glycob/cwac042 ; PMID: 35788318; PMCID: PMC9487896 , Sep-2022